College Physics 2

Magnetic Field

MPS Department | FEU Institute of Technology

College Physics 2

Force on a Moving Charge

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OBJECTIVES

- Analyze magnetic forces on current-carrying conductors
- Explain how current loops behave when placed in a magnetic field
- Explain why currents in the same direction attract while wires carrying opposing currents repel
- Discuss the nature of the force that a moving charged particle experiences in a magnetic field

MAGNETS AND MAGNETIC FIELDS

Any magnet, whether it is in the shape of a bar or a horseshoe, has 2 ends or faces called poles.

The pole of a magnet that points toward the geographic north is called the north pole.

The other pole that points the south is the south pole.

Like poles of a magnet repel while unlike poles attract.

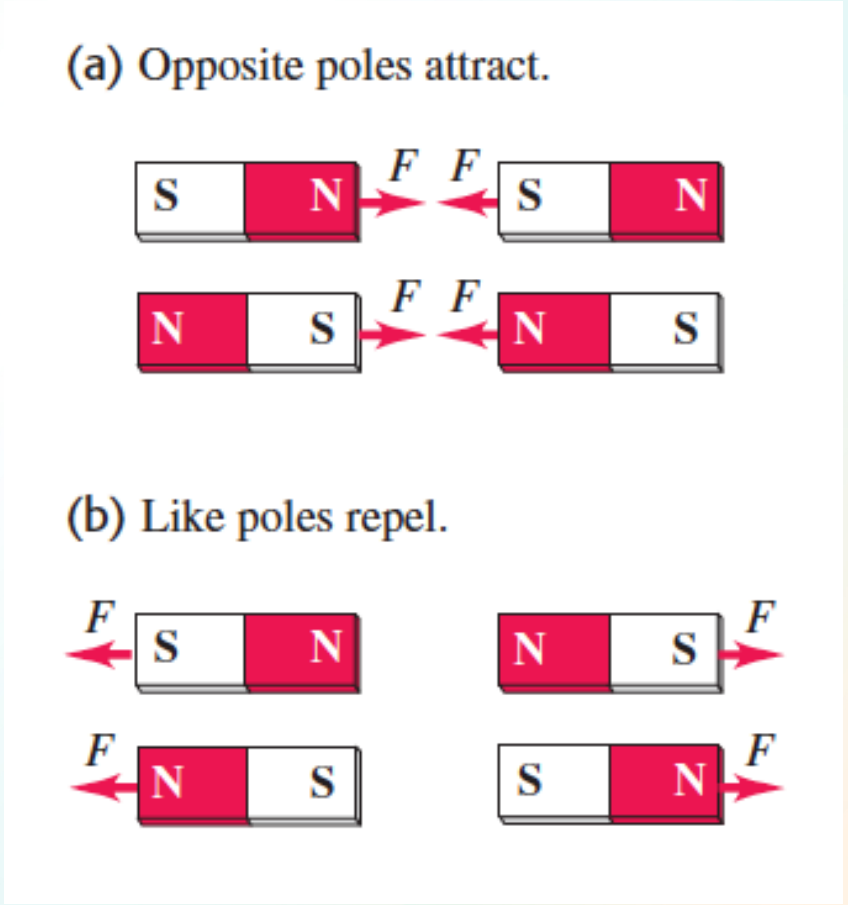


Figure 1 Two bar magnets *a) Opposite poles attracts b) Like poles repel.* From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 884) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

MAGNETS

In the 5th century BC, the Greeks already knew that there are some rocks that attracts bits or iron.

And they are very plentiful in the district of Magnesia and so that's where the name "magnets" and "magnetism" came from. The rocks contains what we call a "magnetite"

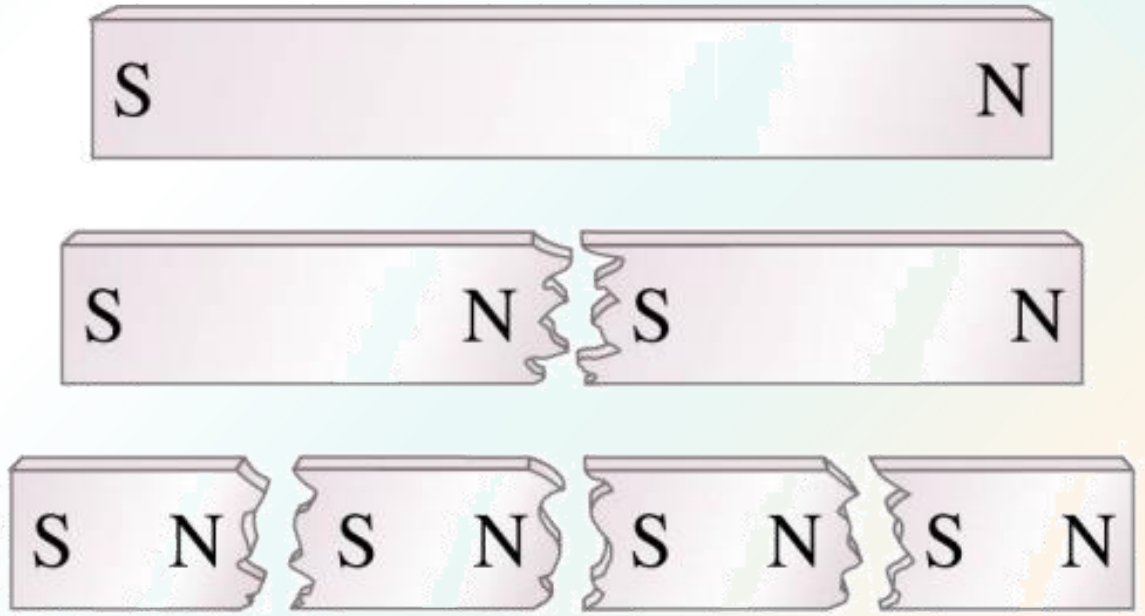
Magnetite has two places of maximum attraction of what we call, POLES.



Figure 2. Magnetite.From. *https://tse4.mm.bing.net/th?id=OIP.35wltD4EAIq_oicACd5gdAHaEo&pid=Api&P=0&w=262&h=164*

MAGNETS AND MAGNETIC FIELDS

If you split a magnet, you won't get isolated north and south poles. New magnets are produced with a north and a south pole.



https://vercauterendavid.files.wordpress.com/2017/12/chapter-27.pdf

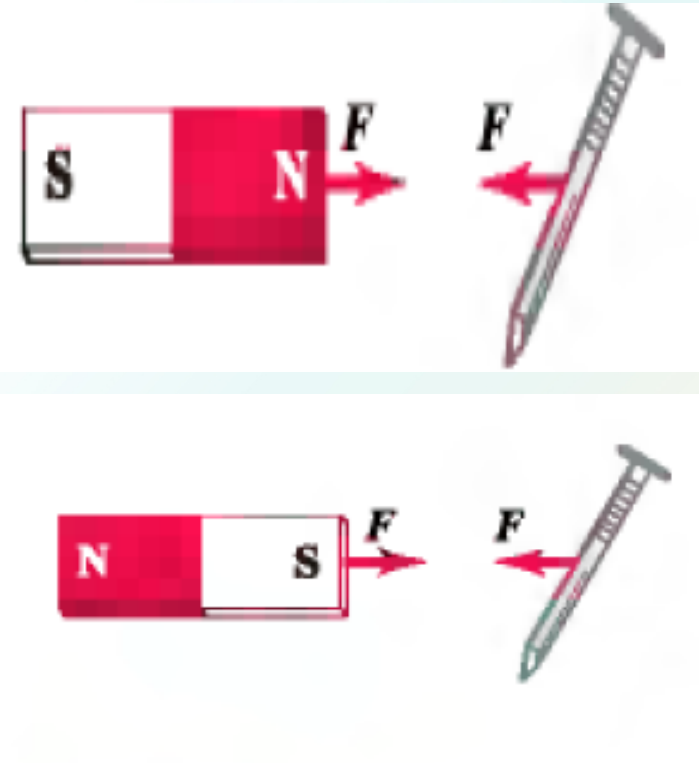
MAGNETS

Permanent magnets (magnetites) were found also to exert forces on pieces of iron that were not magnetized.

It was discovered that when an iron rod is brought in contact with a natural magnet, the rod also becomes magnetized.

When such a rod is suspended by a string from its center, it tends to line up in a north-south direction.

The needle of an ordinary compass is just a piece of *magnetized iron.*



Either pole of a magnet attracts an unmagnetized object that contains iron such as a nail

Figure 4.. Either pole of a magnet attracts an unmagnetized object that contains iron such as a nail. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 884) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

MAGNETS AND MAGNETIC FIELDS

The direction of the magnetic field is tangent to the field line at any point

The number of lines per unit area is proportional to the strength of the magnetic field.

The direction of the magnetic field at a given point can be defined as the direction of that the north pole of a compass needle would point if placed at that point.

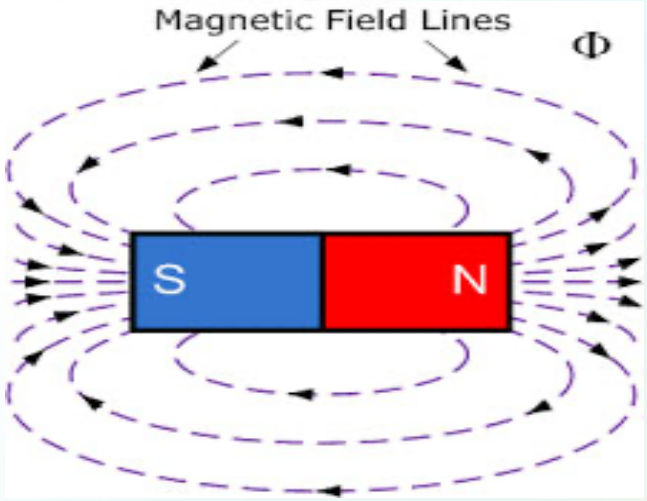


Figure 5. Magnetic Field Lines. From. *https://www.polytechnichub.com/wp-content/uploads/2018/08/magnetic-flux-lines.jpg*



Figure 6. Using compass to show the magnetic field. From. *https://tse2.mm.bing.net/th?id=OIP.JYb8OGDWupgs93Ey2e8L6QAAAA&pid=Api&P=0&w=213&h=177*

EARTH'S MAGNETIC FIELD

The earth acts like a huge magnet.

Its magnetic poles are NOT the same as the geographic poles.

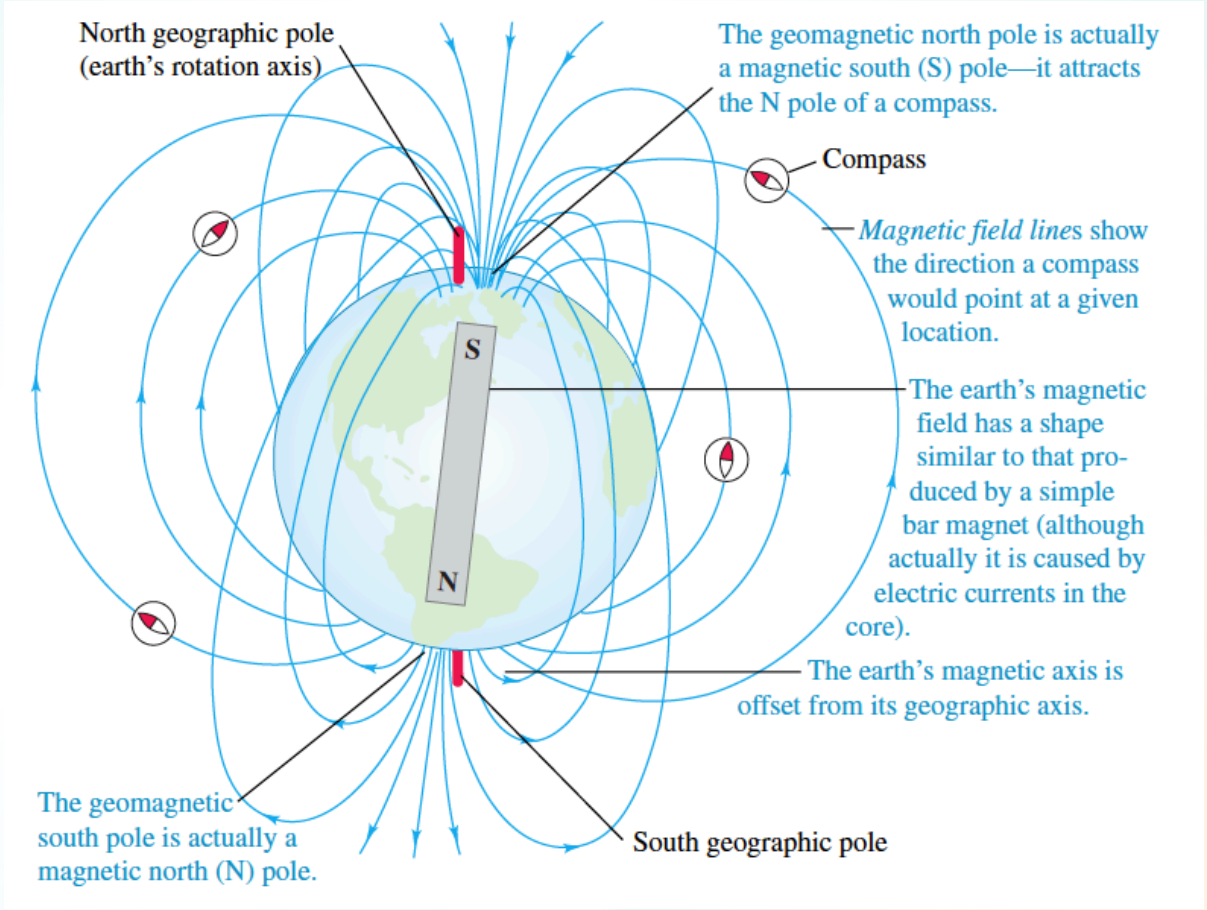


Figure 7.. Earth's magnetic field .. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 884) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

ELECTRIC CURRENTS AND MAGNETIC FIELD





Figure 8. Deflection of compass needles near a current- carrying wire, showing the presence and direction of the magnetic field. From. *https://vercauterendavid.files.wordpress.com/2017/12/chapter-27.pdf*

In 1820, Hans Christian Oersted found that when a compass needle is placed near an electric wire, the needle deflects as soon as the wire is connected to a battery and the wire carries an electric current.

The experiment showed that an electric current produces a magnetic field.

ELECTRIC CURRENTS AND MAGNETIC FIELD

Right hand rule (RHR) is used to determine the direction of the magnetic field (B) relative t the current (I).



Figure 9. *8*agnetic field lines around an electric current in a straight wire . From. *https://vercauterendavid.files.wordpress.com/2017/12/chapter-27.pdf*



Figure 10. Right-hand rule for remembering the direction of the magnetic field . From. *https://vercauterendavid.files.wordpress.com/2017/12/chapter-27.pdf*

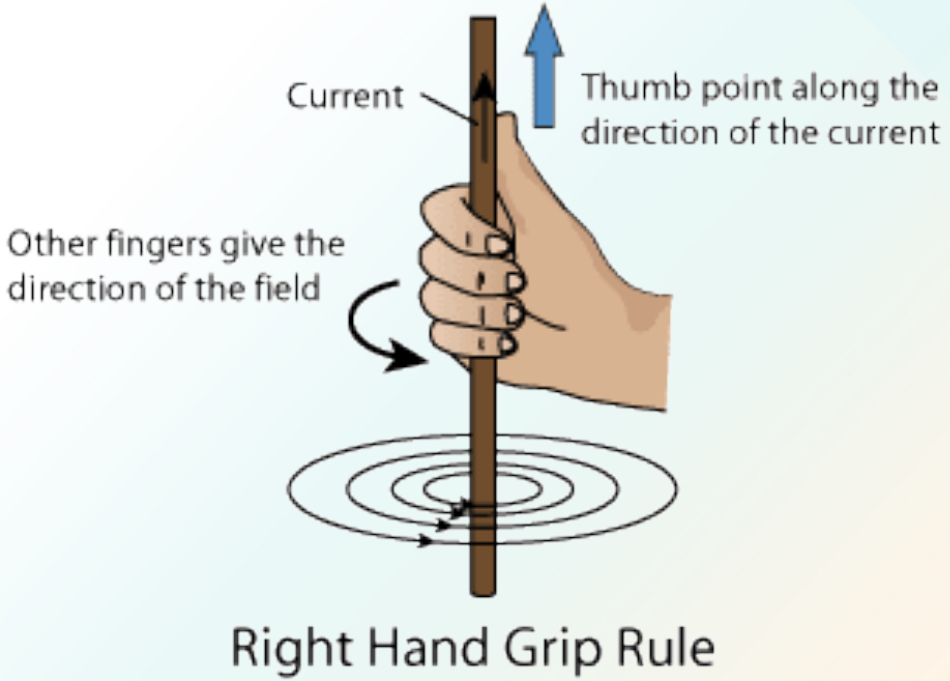


Figure 11. Right hand grip rule. From.. http://bp0.blogger.com/_PyvjcvYfRLk/SFnpDI55a4I/AAAAAAAAAEQ/hiFS_Z2Nwk0/s400/Right-Hand-Grip-Rules.png

FORCE ON ELECTRIC CURRENT IN MAGNETIC FIELD

Magnet exerts a force on a current-carrying wire. Direction of force is determined by right-hand rule.

$$F = IlB\sin\theta$$

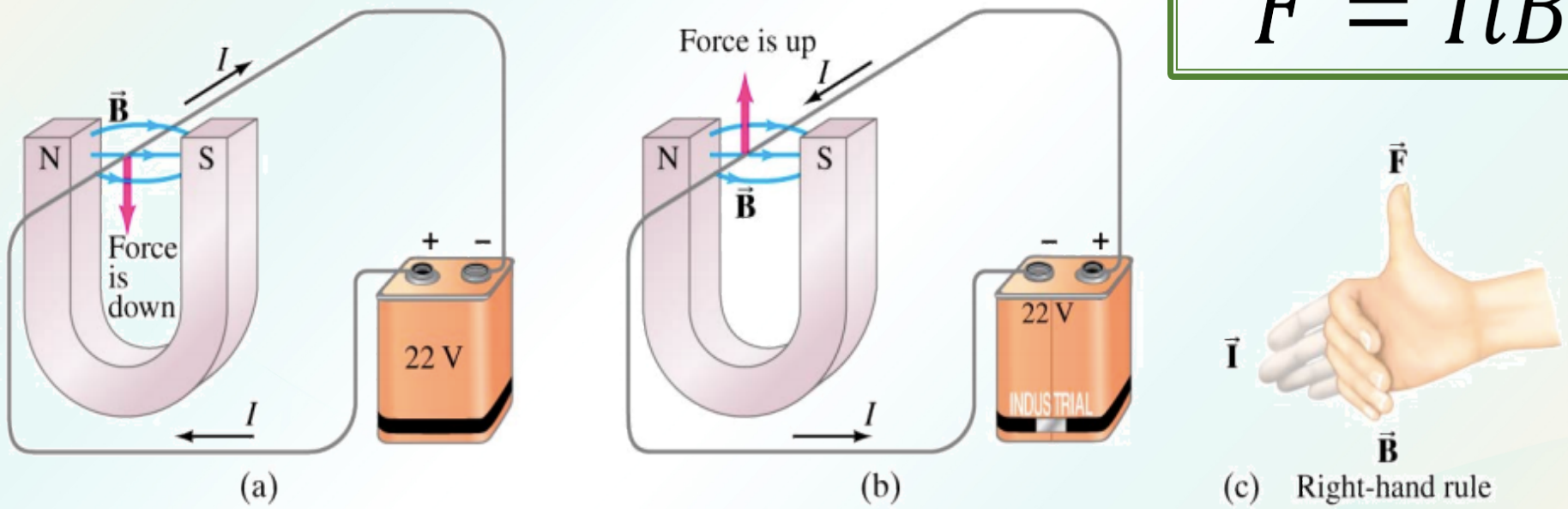


*Figure 12. a)*Force on an electric current in a magnetic field .b) same, but current reversed; c) right-hand rule for setup in b). From. *https://vercauterendavid.files.wordpress.com/2017/12/chapter-27.pdf*

FORCE ON ELECTRIC CURRENT IN MAGNETIC FIELD

$$F = I\ell B\sin\theta$$

$$F_{max} = I\ell B$$

Maximum force occurs when $\theta = 90^0$.
Force is zero if wire is parallel to field lines.

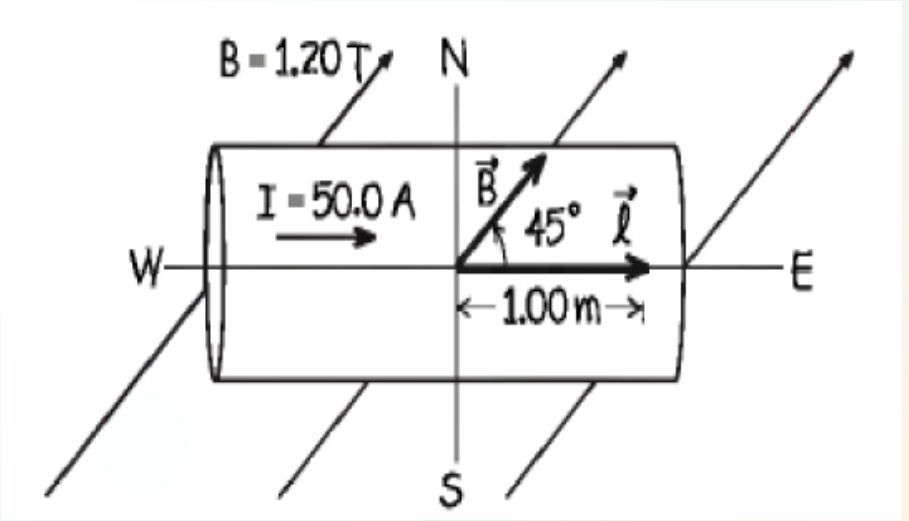
Where F = Force (N)
I = current (A)
l = length of wire exposed to magnetic field (m)
B = magnetic field, tesla (T),
1T = 1N/Am = 1 weber per meter squared =1 Wb/m^2 = 10^4 gauss = 10^4G
θ =angle between the current (I) direction and the magnetic field (B)

The force on the wire depends on the current, the length of wire, the magnetic field and the orientation. The magnetic force is perpendicular to the magnetic field direction.

Sample Problem

1. A straight horizontal copper rod carries current of 5.0 A from west to east in a region between the poles of an electromagnet. In this region there is a horizontal magnetic field toward the north east (45^0 north of east) with magnitude 1.20 T. Find the magnitude of the force on the wire with a length of 1.0 m rod.

Answer: 4.243 N



Characteristics of the Magnetic Force on a Moving Charge

There are four key characteristics of the magnetic force on a moving charge.

1. The magnitude of the magnetic force is proportional to the magnitude of the charge and to the speed v of the particle.
2. The magnitude of force is also proportional to the magnitude or "strength" of the magnetic field.
3. The magnetic force is proportional to sinΘ.
4. The magnetic force is always perpendicular to the magnetic field and the velocity, that is, perpendicular to the plane formed by $\vec{v}$ *and* $\vec{B}$

FORCE ON ELECTRIC CHARGE MOVING IN A MAGNETIC FIELD

The force on charged particles due to a magnetic field is perpendicular to the magnetic field direction.

Right-hand rule is used to determine the direction of force. It is only true for positively charged particles. For negatively charged particle, the force is the opposite direction.

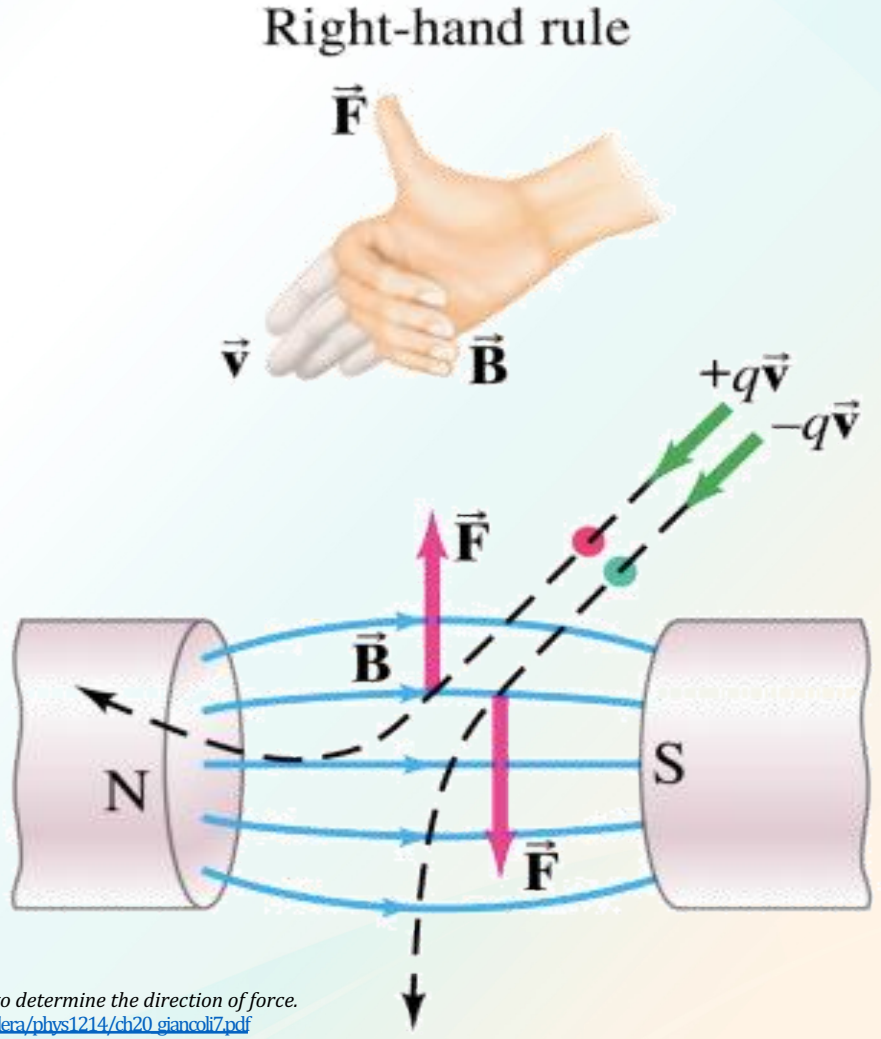


Figure 13.. Right hand rule to determine the direction of force.
From. http://hep0.okstate.edu/flera/phys1214/ch20_giancoli7.pdf

FORCE ON ELECTRIC CHARGE MOVING IN A MAGNETIC FIELD

$$F = qvB\sin\theta$$

$$F_{max} = qvB$$

Maximum force occurs when $\theta = 90^0$.
Force is zero if particle moves parallel to field lines.

Where

F = Force (N)
q = charge (C)
v = velocity of charge (m/s)
B = magnetic field, tesla (T),

1T = 1N/Am = 1 weber per meter squared =1 Wb/m^2 = 10^4 gauss = 10^4 G

θ =angle between the current (I) direction and the magnetic field (B)

The force on the wire depends on the current, the length of wire, the magnetic field and the orientation. The magnetic force is perpendicular to the magnetic field direction.

SUMMARY OF RIGHT HAND RULES

TABLE 20–1 Summary of Right-hand Rules (= RHR)

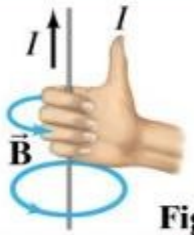
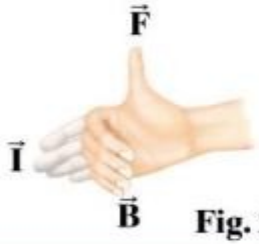
Physical Situation	Example	How to Orient Right Hand	Result
1. Magnetic field produced by current (RHR-1)	 Fig. 20–8d	Wrap fingers around wire with thumb pointing in direction of current I	Fingers curl in direction of $\vec{\mathbf{B}}$
2. Force on electric current I due to magnetic field (RHR-2)	 Fig. 20–11c	Fingers first point straight along current I, then bend along magnetic field $\vec{\mathbf{B}}$	Thumb points in direction of the force $\vec{\mathbf{F}}$
3. Force on electric charge $+q$ due to magnetic field (RHR-3)	 Fig. 20–15	Fingers point along particle's velocity $\vec{\mathbf{v}}$, then along $\vec{\mathbf{B}}$	Thumb points in direction of the force $\vec{\mathbf{F}}$

Table 1. Summary of Right-hand rule. From. *http://hep0.okstate.edu/flera/phys1214/ch20_giancoli7.pdf*

Alternative Rule

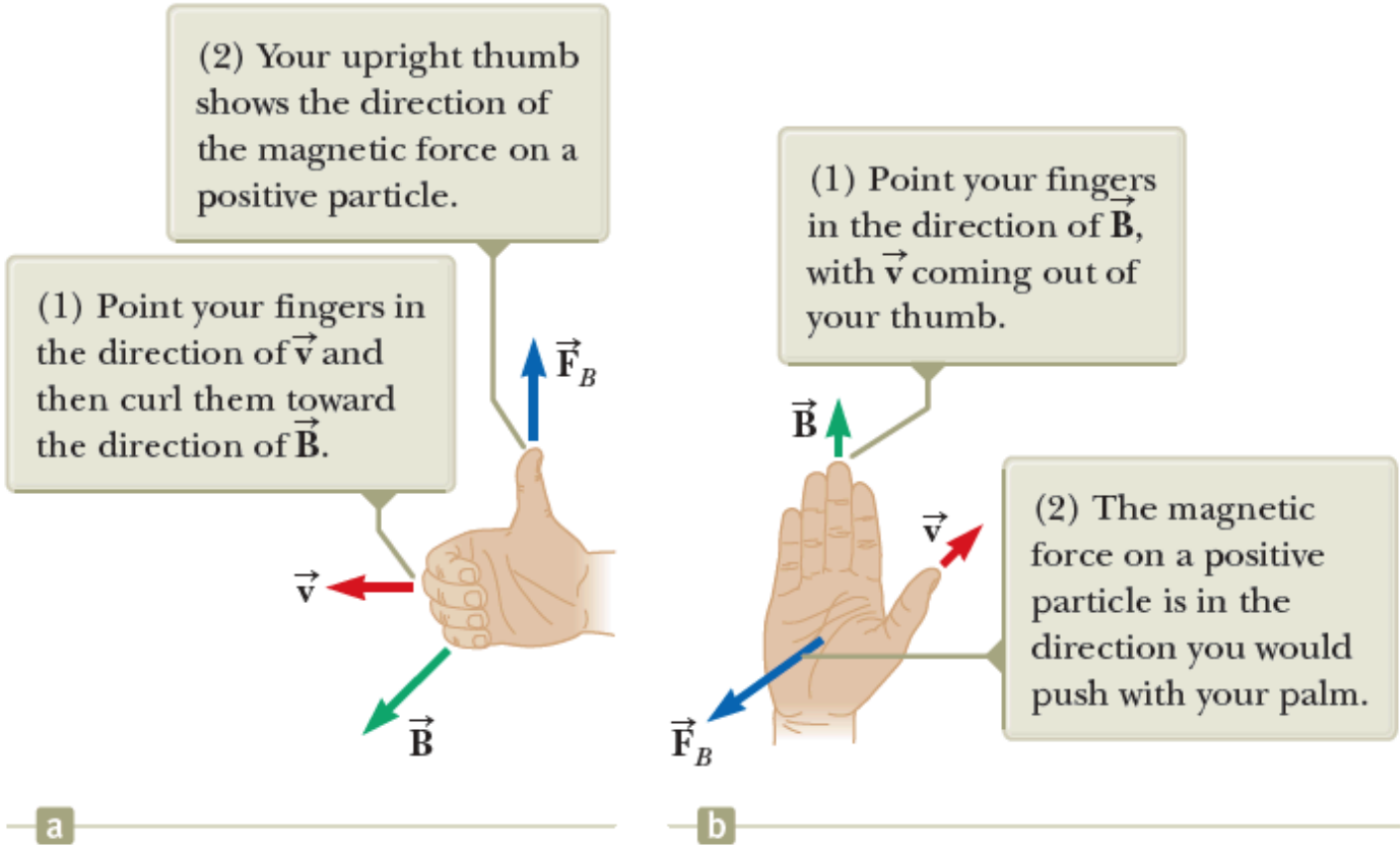


Figure 29.5 Two right-hand rules for determining the direction of the magnetic force $\vec{F}_B = q\vec{v} \times \vec{B}$ acting on a particle with charge q moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$. (a) In this rule, the magnetic force is in the direction in which your thumb points. (b) In this rule, the magnetic force is in the direction of your palm, as if you are pushing the particle with your hand.

Figure 13. Alternative right-hand rule. From. *Physics for Scientists and Engineers with Modern Physics*(pp 872) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

Sample Problem

An electron in an old-style television picture tube moves toward the front of the tube with a speed of 8.0 x 10^6 m/s along the x axis . Surrounding the neck of the tube are coils of wire that create a magnetic field of magnitude 0.025 T, directed at an angle of 60° to the x axis and lying in the xy plane. Calculate the magnetic force on the electron.

Answer: 2.8 x 10^{-14} N

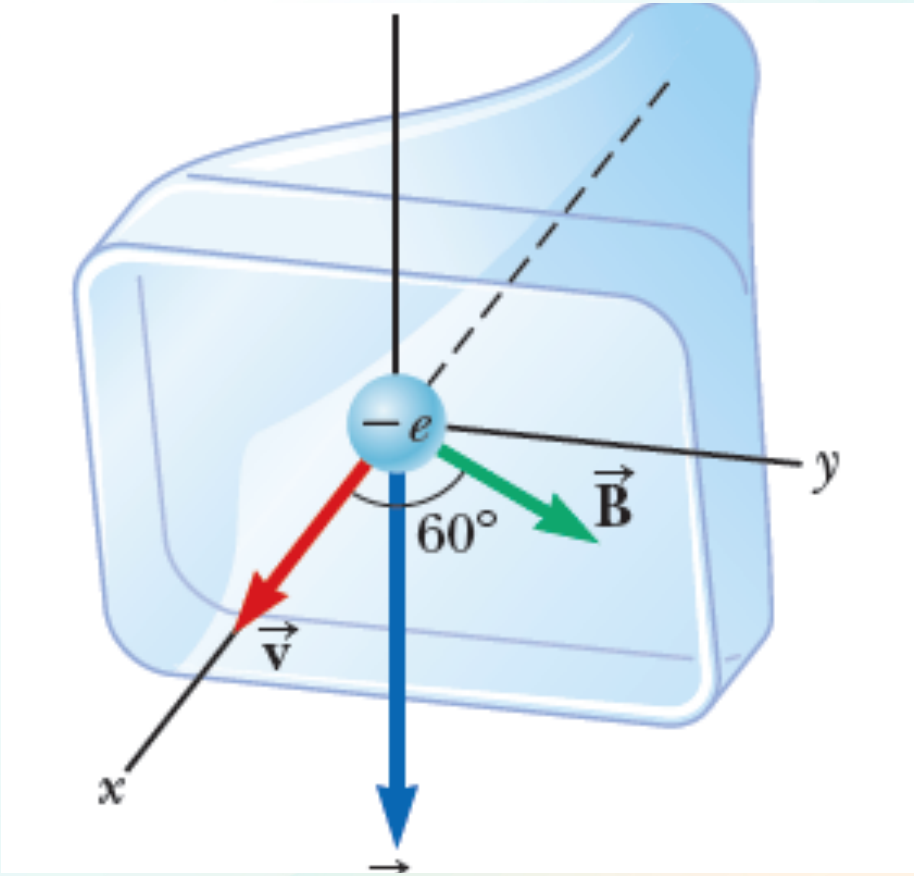


Figure 14. Electron moving in an electric field. From. *Physics for Scientists and Engineers with Modern Physics*(pp 874) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

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https://tse4.mm.bing.net/th?id=OIP.35wltD4EAIq_oicACd5gdAHaEo&pid=Api&P=0&w=262&h=164

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https://www.slideserve.com/Jims/

http://hep0.okstate.edu/flera/phys1214/ch20_giancoli7.pdf

College Physics 2

Torque on a Current Loop

MPS Department | FEU Institute of Technology

OBJECTIVES

- Explain how current loops behave when placed in a magnetic field
- Solve problems involving magnetic field.
- Explain why currents in the same direction attract while wires carrying opposing currents repel
- Discuss Ampere's Law and what it tells about magnetic fields.
- Use Ampere's Law to calculate the magnetic field of symmetric current distributions.

The Torque on a Current-Carrying Coil

The two forces on the loop have equal magnitude but an application of RHR-1 shows that they are opposite in direction.

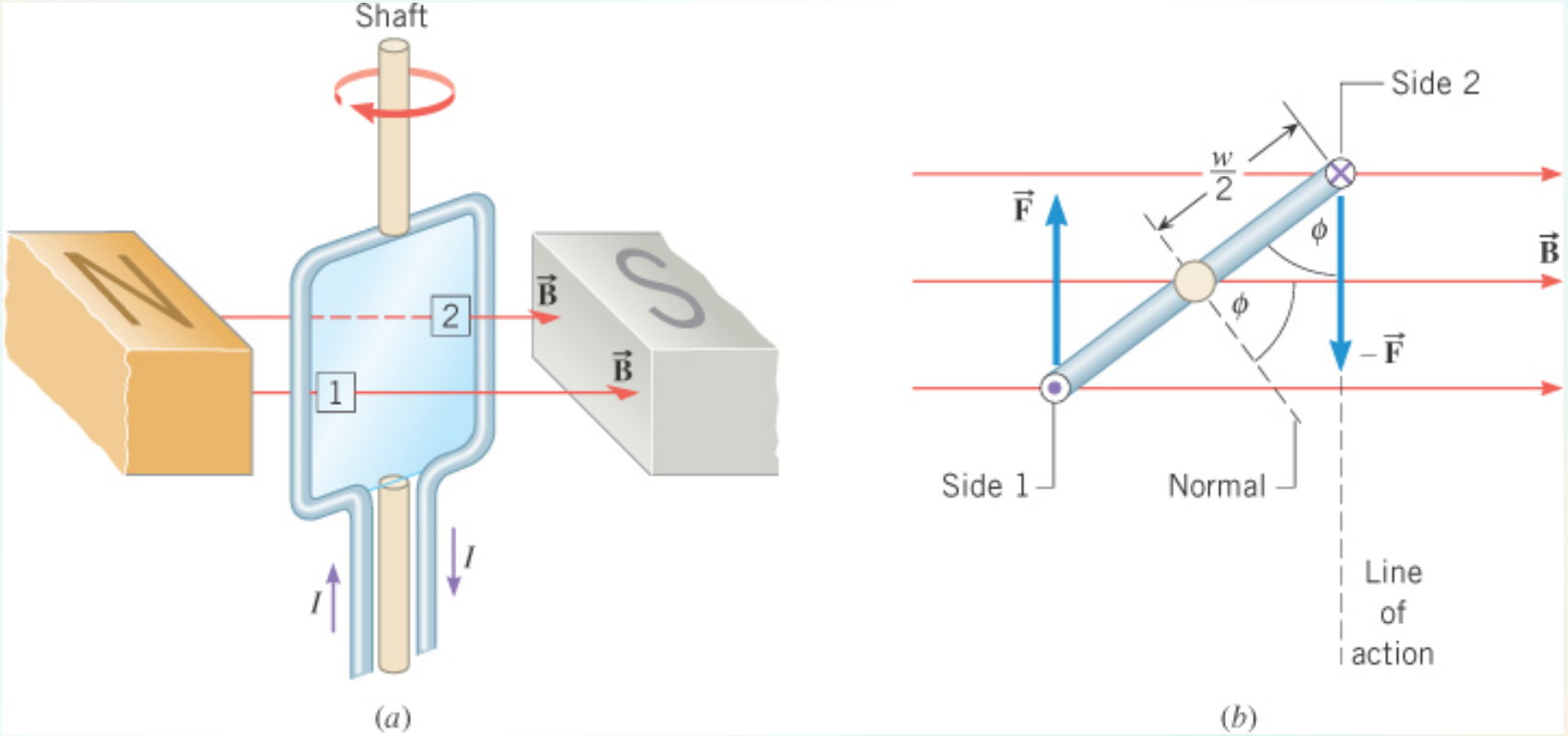


Figure 15. Torque on a current-carrying coil. From. *https://image.slidesharecdn.com/ch21magneticfieldsandforces23july2015-161130010302/95/ch21-magnetic-fields-and-forces-23-july-2015-39-638.jpg?cb=1480467828*

Torque on Current Loop

The forces on opposite sides of a current loop will be equal and opposite (if the field is uniform and the loop is symmetric) but there may be a torque.

$$\tau = NIABsin\theta$$

$$M = NIA$$

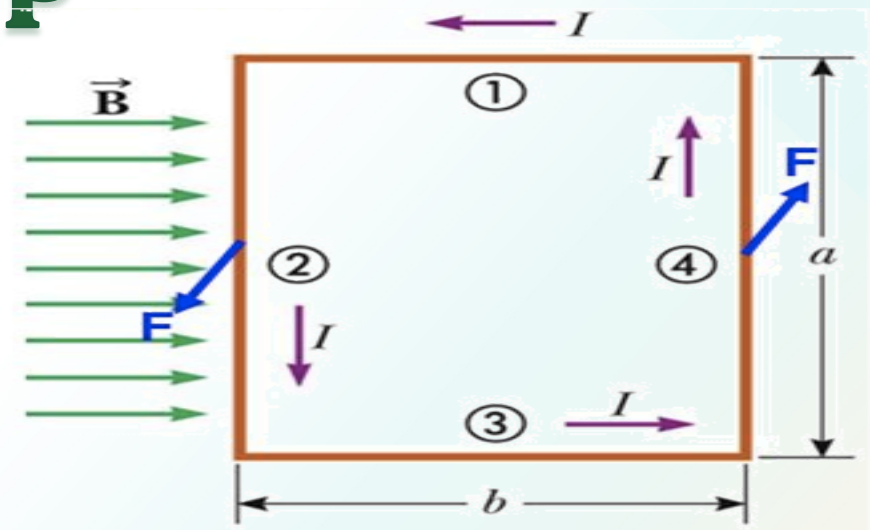
Torque on Current Loop

The rectangular loop carries a current I in a uniform magnetic field.

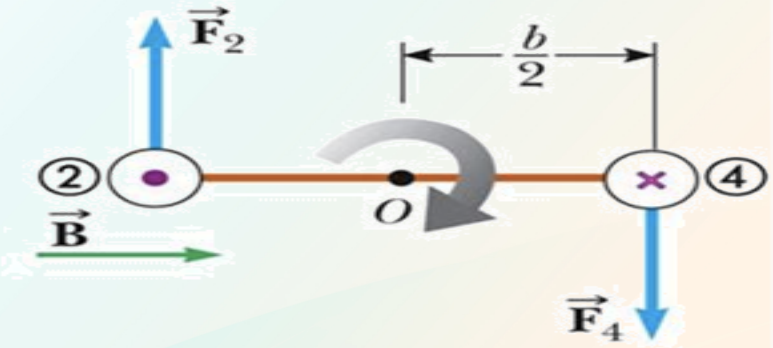
No magnetic force acts on sides 1 and 3

Magnitude of the magnetic force on sides 2 & 4, F = IAB

The direction of F2 is out of the page.
The direction of F4 is into the page.



Front view



bottom view

Sample Problem

A circular coil of wire has a diameter of 20.0 cm and contains 10 loops. The current in each loop is 3.00 A, and the coil is placed in a 2.00 T external magnetic field. Determine the maximum and minimum torque exerted on the coil by the field.

Solution:

$$\tau = NIABsin\theta$$

$$A = \pi r^2, \quad \text{r = d/2}, \quad \text{max T if } \theta = 90^0$$

$$\tau_{max} = (10)(\pi)(0.1m)^2(2T)(\sin 90)$$

$$\boldsymbol{\tau_{max} = 1.88\ Nm}$$

$$\boldsymbol{\tau_{min} = 0}\ , when\ \theta = 0^0$$

APPLICATIONS

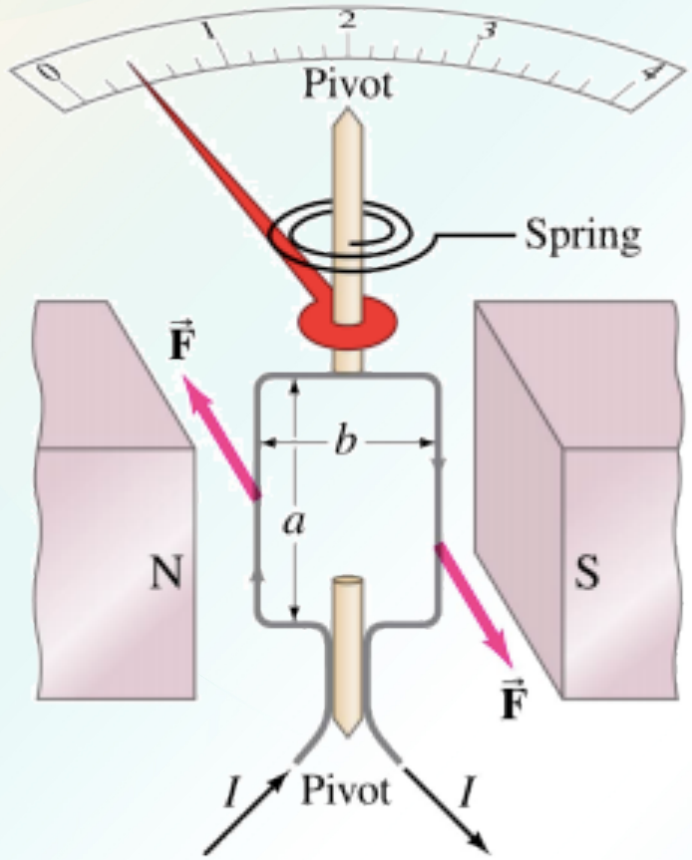


Figure 16.. Galvanometer . From. http://hep0.okstate.edu/flera/phys1214/ch20 giancoli7.pdf

A galvanometer consists of a coil of wire (with attached pointer) suspended in the magnetic field of a permanent magnet. When the current flows through the loop of wire, the magnetic field exerts a torque on the loop.

Galvanometer is the basic component of analog meters including ammeters, voltmeters and ohmmeters.

APPLICATIONS

An electric motor changes electric energy into rotational mechanical energy. It works on the same principle as the galvanometer, except that there is no spring so the coil can rotate continuously in one direction.

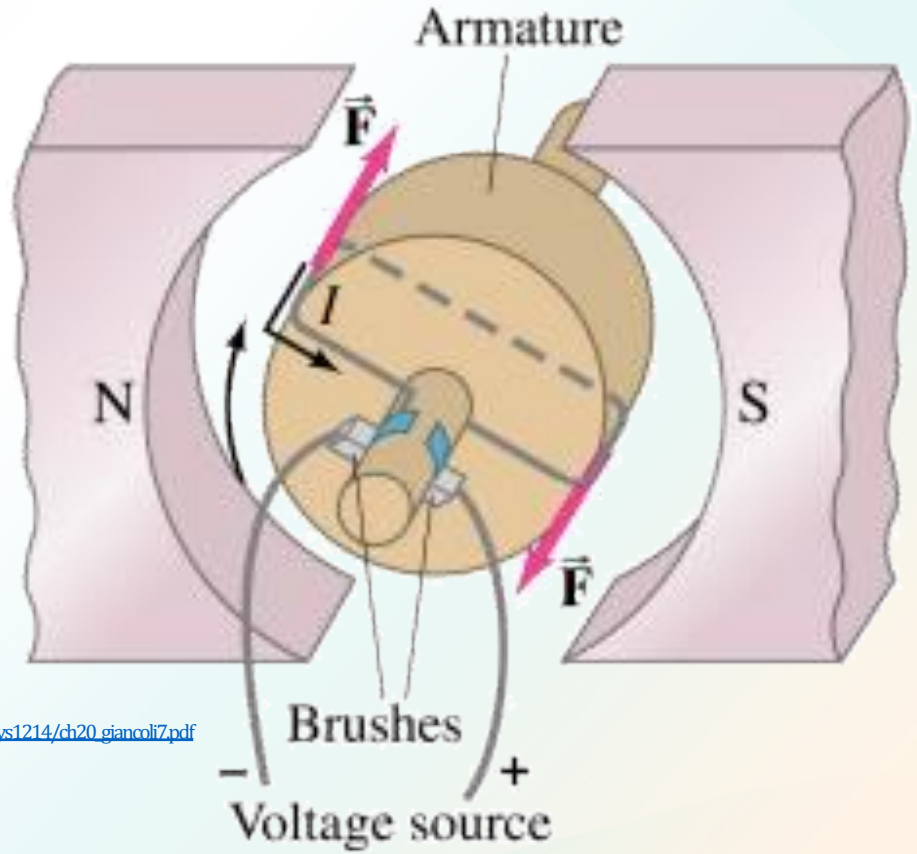


Figure 17. Right Simple DC motor. From. http://hep0.okstate.edu/flera/phys1214/ch20_giancoli7.pdf

Diagram of a simple dc motor.

APPLICATIONS

A loudspeaker works on the principle that a magnet exerts a force on a current-carrying wire. The electrical output of a stereo or TV set is connected to the wire leads of the speaker. The speaker leads are connected internally to a coil of wire, which is itself attached to the speaker cone.

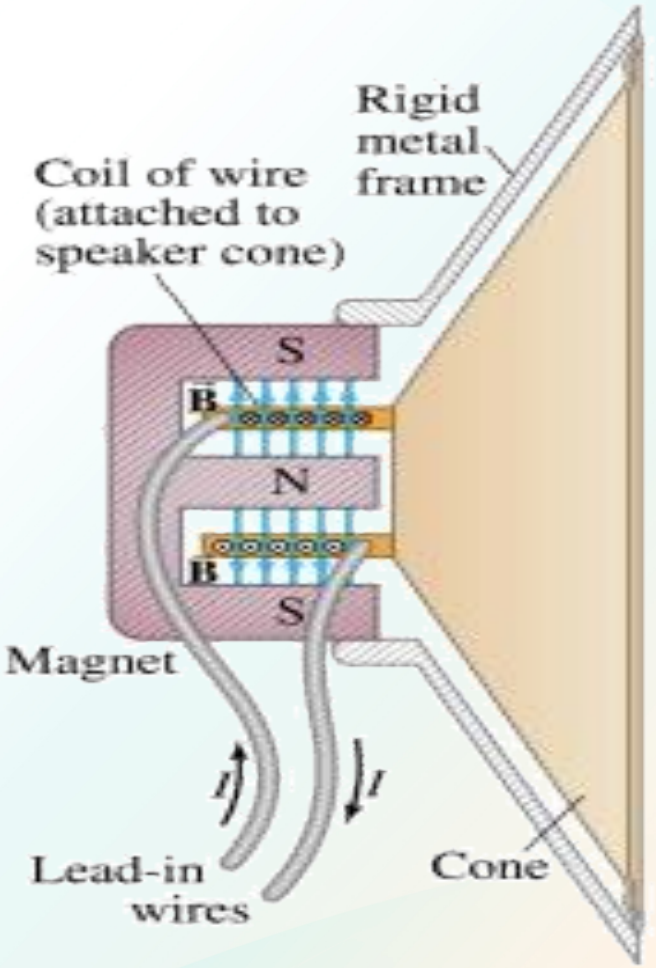


Figure 18.. Loudspeaker. From. http://hep0.okstate.edu/flera/phys1214/ch20_giancoli7.pdf

MAGNETIC FIELD DUE TO LONG STRAIGHT WIRE

$$B = \frac{\mu_0 I}{2\pi r}$$

Where B = magnetic field, tesla (T),
I = current, A
r = distance from the wire (m)
μ_0 = permeability of free space (Tm/A)
$\mu_0 = 4\pi \times 10^{-7}$ Tm/A

Magnetic field is inversely proportional to the distance from the wire.

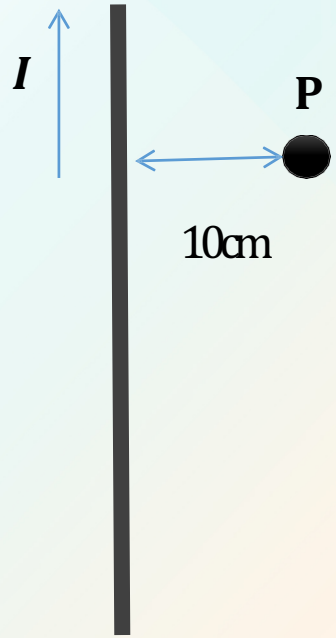
Sample Problem

1. An electric wire in the wall of a building carries a DC current of 25 A vertically upward. What is the magnetic field due to this current at a point P 10 cm due north of the wire?

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \left(\frac{4\pi x 10^{-7} Tm/A)(25A)}{4\pi(0.10m)}\right.$$

$$B = 5x10^{-5}T$$



Direction can be determined using right hand rule. Direction of B is to the west into the page.

FORCE BETWEEN TWO PARALLEL WIRES

The magnetic field produced at the position of wire 2 due to the current wire 1:

$$B_1 = \frac{\mu_0}{2\pi}\frac{I_1}{d}$$

The force this field exerts on a length of wire 2 is:

$$F_2 = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d} l_2$$

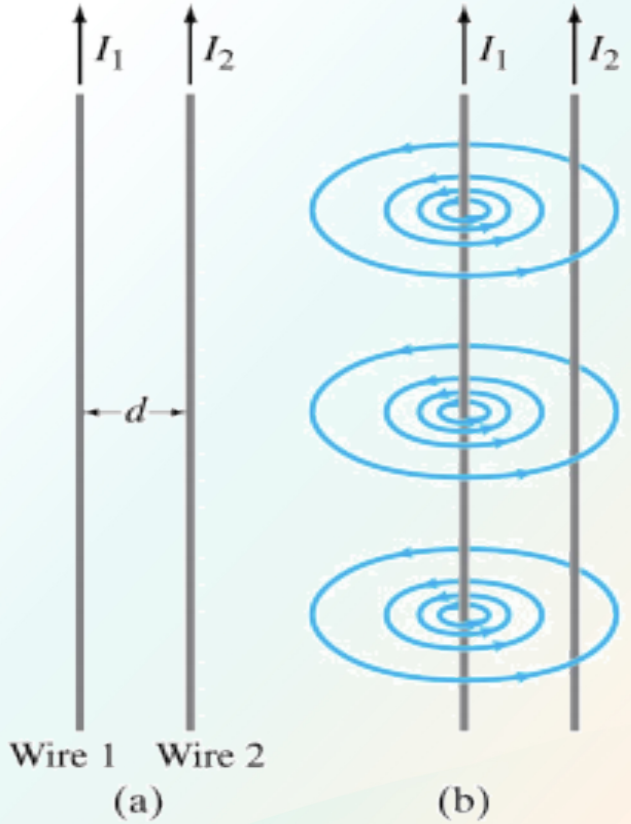


Figure 19. a) two parallel conductors carrying currents, I1 and I2, b) the magnetic field B1 produced by I1. B1 points into page at position of I2. From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

FORCE BETWEEN TWO PARALLEL WIRES

Parallel currents in the same direction exert an attractive force on each other.

Antiparallel currents (opposite directions) exert a repulsive force on each other.

$$B_1 = \frac{\mu_0}{2\pi}\frac{I_1}{r}$$

$$F_2 = I_2 l_2 B_1$$

$$F_2 = I_2 l_2 \left(\frac{\mu_0}{2\pi}\frac{I_1}{r}\right) = \left(\frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}\right) l_2$$

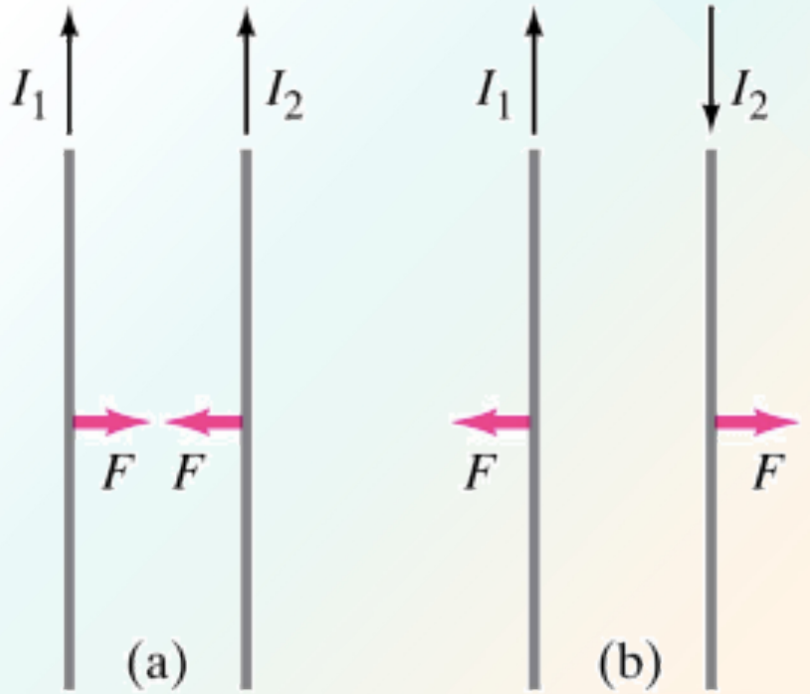


Figure 20. a) parallel currents in the same direction, b) antiparallel currents . From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

Sample Problem

2. The 2 wires of a 2m long appliance cord are 3.0 mm apart and carry a current of 8.0 A dc. Calculate the force one wire exerts on the other.

$$F_2 = \left(\frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}\right) l_2$$

$$F_2 = \left(\frac{4\pi x 10^{-7} Tm/A)(8A)(8A)(2m)}{2\pi(3x10^{-3}m)}\right)$$

$$\boldsymbol{F_2 = 8.5\, x10^{-3} N}$$

Solenoids and Electromagnets

A solenoid is a long coil of wire consisting of many loops or turns. If tightly wrapped, the magnetic field in its interior is almost uniform.

It acts like a magnet. One end can be considered the north pole while the other end is the south pole depending on the direction of current in the loops.

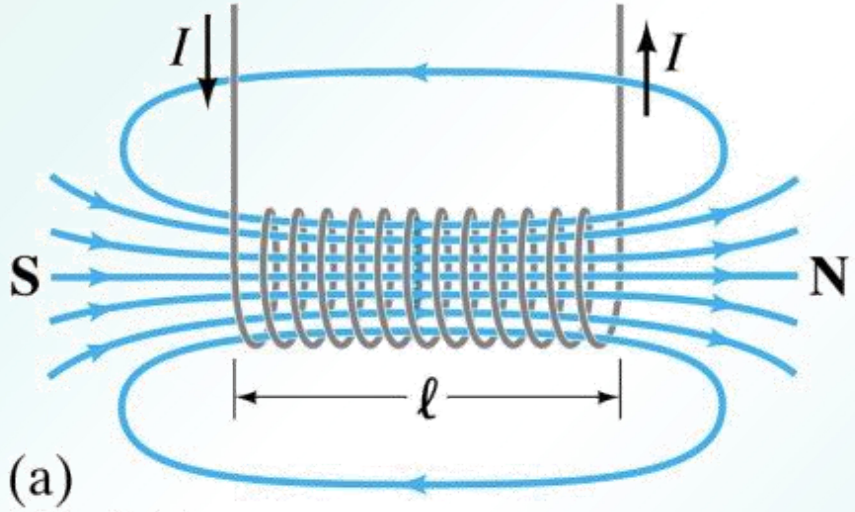


Figure 21. a) Solenoid . From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

$$B = \frac{\mu_0 N I}{l}$$

Where N = number of turns
I = current
l =length of wire

SOLENOIDS AND ELECTROMAGNETS

If a piece of iron is place inside a solenoid, the magnetic field is increased greatly because the iron becomes a magnet. The magnetic field is the sum of that due to current and that due to iron and it can be hundreds or thousands of times that due to the current. Such an an iron-core solenoid is called electromagnet.

Applications include motors and generators. One simple application of electromagnet is doorbell.

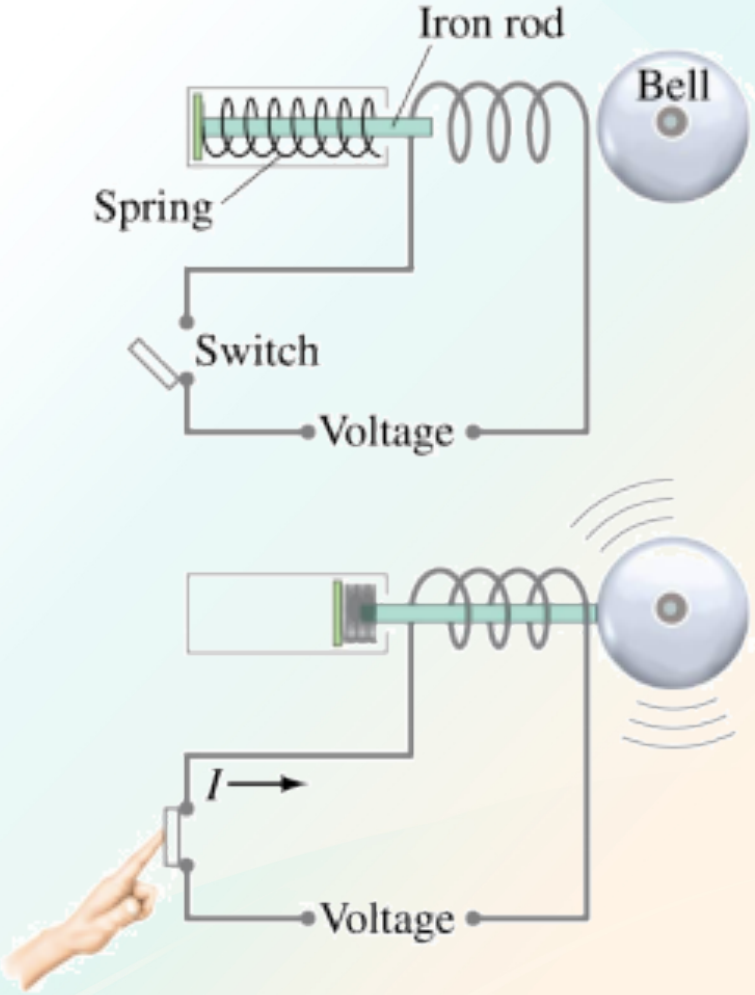


Figure 22. Iron-core solenoid . From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

Sample Problem

3. A thin 12 cm long solenoid has a total of 420 turns of wire and carries a current of 2.0A. Calculate the field inside near the center.

$$B = \frac{\mu_0 NI}{l}$$

$$B = \frac{(4\pi x 10^{-7} Tm/A)(420)(2A)}{0.12m}$$

$$\boldsymbol{B = 8.80\ x10^{-3}T}$$

AMPERE'S LAW

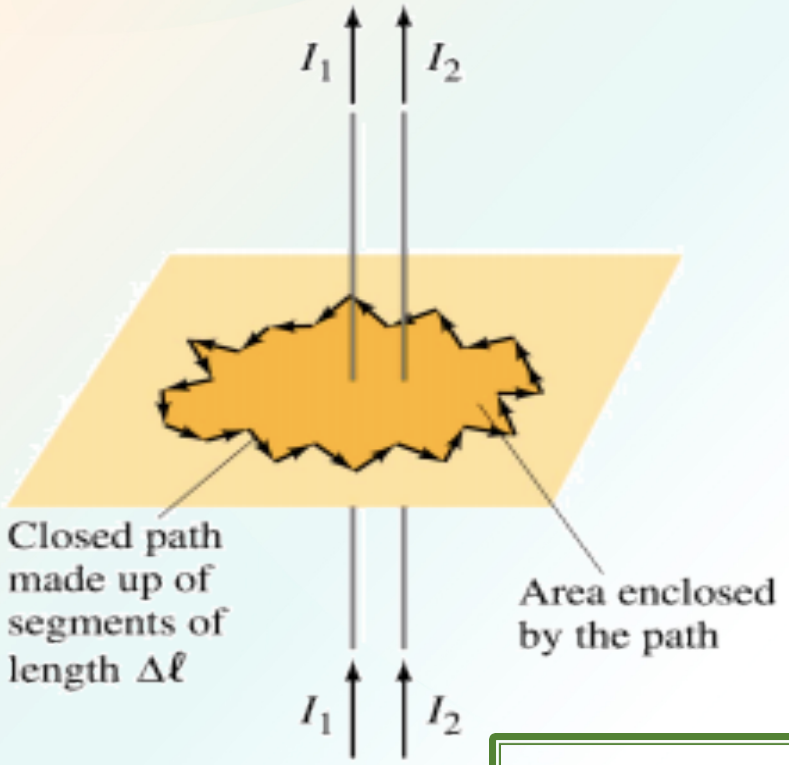


Figure 23. Ampere's Law. From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

$$\Sigma B_{II} \Delta\ell = \mu_0 I_{encl}$$

The sum of all the products of the the length of each segment and the component of magnetic field parallel to the segment is equal to the u0 times the current that passes through the surface enclosed by the path.

AMPERE'S LAW

Field due to straight wire:

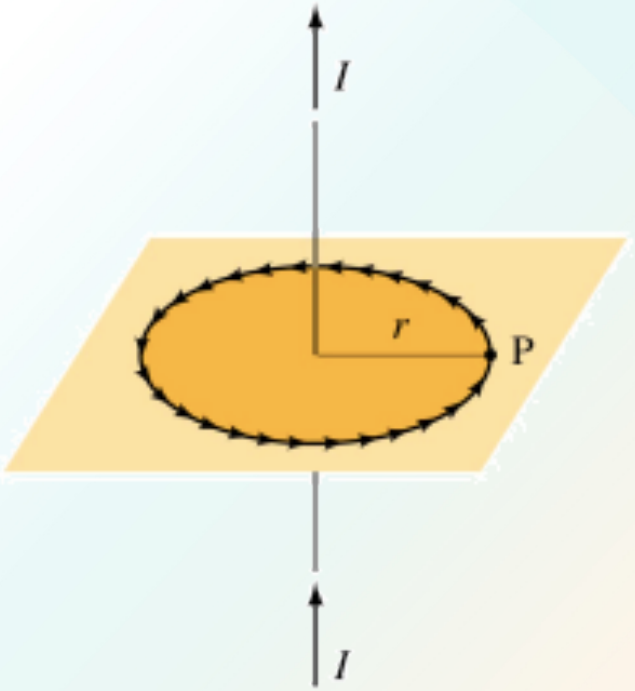
$$\Sigma B_{II}\Delta\ell = \mu_0 I_{encl}$$

$$B_1\Delta l_1 + B_2\Delta l_2 + \ldots + B_{100}\Delta l_{100} = \mu_0 I$$

$$B(\Delta l_1 + \Delta l_2 + \cdots + \Delta l_{100}) = \mu_0 I$$

$$B \cdot 2\pi r = \mu_0 I$$

$$\boldsymbol{B = \frac{\mu_0 I}{2\pi r}}$$



Because of symmetry, all B_i's are the same.

Figure 24. Field due to straight wire. From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

AMPERE'S LAW

Field inside a solenoid:

$$\Sigma B_{II}\Delta\ell = \mu_0 I_{encl}$$

$$(B_{II}\Delta l)_{ab} + (B_{II}\Delta l)_{bc} + (B_{II}\Delta l)_{cd} + (B_{II}\Delta l)_{da} = \mu_0 I_{enc}$$

$$B \text{ at } ab = 0;$$

$$\text{Since } B \text{ is perpendicular to segment,}$$

$$B \text{ at } bc \text{ and } da = 0$$

$$(B_{II}\Delta l)_{cd} = \mu_0 I_{enc}$$

$$B\ell = \mu_0 NI$$

$$\boldsymbol{B = \frac{\mu_0 NI}{\ell}}$$

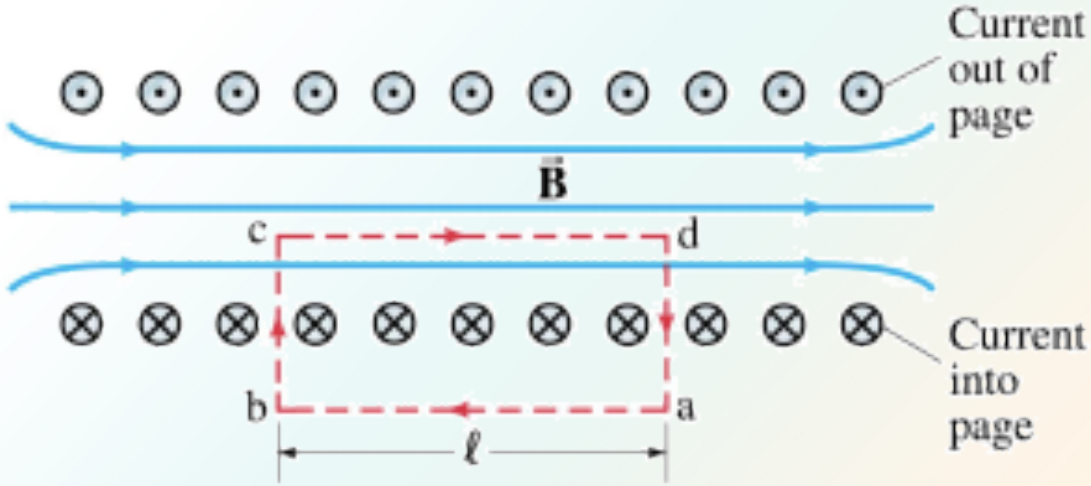


Figure 25. Field inside a solenoid. From. *https://image.slidesharecdn.com/ppa6lecturech20-100424010240-phpapp02/95/ppa6-lecture-ch20-20-728.jpg?cb=1272071064*

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